In quantum mechanics, the situation is fundamentally different. *Any* interaction that is strong enough to measure some aspect of a system is necessarily strong enough to disrupt some other aspect of the same system. Thus, you can learn nothing about a quantum system without changing something else.

This should be evident in the examples involving $\mathcal{A}$ and σ. Suppose we begin with $\sigma = +1$ along the z axis. If we measure σ again with $\mathcal{A}$ oriented along z, we will confirm the previous value. We can do this over and over without changing the result. But consider this possibility: Between successive measurements along the z axis, we turn $\mathcal{A}$ through 90 degrees, make an intermediate measurement, and turn it back to its original direction. Will a subsequent measurement along the z axis confirm the original measurement? The answer is no. The intermediate measurement along the x axis will leave the spin in a completely random configuration as far as the next measurement is concerned. There is no way to make the intermediate determination of the spin without completely disrupting the final measurement. One might say that measuring one component of the spin destroys the information about another component. In fact, one simply cannot simultaneously know the components of the spin along two different axes, not in a reproducible way in any case. There is something fundamentally different about the state of a quantum system and the state of a classical system.

1.5 PROPOSITIONS

The space of states of a classical system is a mathematical set. If the system is a coin, the space of states is a set of two elements, *H* and *T*. Using set notation, we would write {*H, T*}. If the system is a six-sided die, the space of states has six elements labeled {1, 2, 3, 4, 5, 6}. The logic of set theory is called *Boolean* logic. Boolean logic is just a formalized version of the familiar classical logic of propositions.

A fundamental idea in Boolean logic is the notion of a truth-value. The truth-value of a proposition is either *true* or *false*. Nothing in between is